

Professional Agentic AI Engineering

20 Sessions · 80 Hours · From Python to Production

Programme Objectives

10 things you will be able to do when you finish — actual skills you can demonstrate on Day 1 after you leave.

01	Build multi-agent AI systems where specialised agents collaborate to solve complex, real-world tasks autonomously
02	Design and implement end-to-end RAG pipelines - from document ingestion to production-grade retrieval and generation
03	Apply advanced retrieval techniques: semantic chunking, hybrid search, reranking, query rewriting and adaptive routing
04	Develop production-ready LLM applications using LangChain, LangGraph, CrewAI, AutoGen and Azure AI platforms
05	Integrate AI agents with external tools, APIs, databases and real-time data sources via MCP and function calling
06	Implement short-term and long-term memory in agents and orchestrate multi-agent workflows using LangGraph state machines
07	Evaluate and debug AI agent systems using LangSmith tracing, LLM-as-a-judge, hallucination detection and guardrails
08	Optimise AI systems for latency, cost and performance - caching, retrieval tuning, model selection strategy
09	Deploy and monitor scalable AI applications with FastAPI, Streamlit, Hugging Face and observability tools
10	Build safe, responsible AI systems with guardrails, privacy controls and ethical governance frameworks

The Curriculum

20 sessions · 80 hours · Each session is 4 hours of live, hands-on training.

#	Session	What You Build / Learn	Tools
1	Python Setup & Foundations for AI	Python environments (venv/conda), VS Code setup, pip, variables, data types, strings, loops, functions. Build a simple prompt formatter.	<i>Python</i> · VS Code
2	Python for Data & APIs	Lists, dicts, JSON, file I/O, try-except, requests library, API keys, environment variables. Call a live API.	<i>requests</i> · <i>json</i> · <i>os</i>

#	Session	What You Build / Learn	Tools
3	OOP, Async & Data Handling for AI Systems	Classes for AI components, async/await for parallel API calls, pandas basics, regex text processing. Mini AI pipeline.	<i>pandas · asyncio</i>
4	NLP Foundations & Text Processing	Tokenization, lemmatization, embeddings, Word2Vec, TF-IDF, classification, sentiment, NER, POS, ingestion with pandas.	<i>NLTK · SpaCy</i>
5	Language Models & Transformers	RNN to LSTM to Transformers evolution, attention, BPE/WordPiece, pretraining vs fine-tuning, RLHF, LLMs vs SLMs.	<i>TensorFlow · HuggingFace</i>
6	Generative AI, Prompting & Multi-Modal	Prompt engineering, temperature/top-k/top-p, hallucination control, multi-modal (text+vision+audio), GPT vision, voice AI, STT.	<i>Azure AI · Copilot</i>
7	AI Platforms & LangChain Foundations	OpenAI API, LangChain - chat models, prompt templates, chains, tools, memory. AutoGen intro. Build quiz bot and chatbot.	<i>LangChain · AutoGen</i>
8	Vector Databases & Introduction to RAG	Embeddings, vector DBs (ChromaDB, FAISS, Pinecone), similarity search, document loaders, chunking. Build a retrieval pipeline.	<i>ChromaDB · LangChain</i>
9	Production RAG	Document ingestion, chunking, query transformation, hybrid retrieval, context optimisation, caching, RAGAS evaluation.	<i>LangChain · Azure OpenAI</i>
10	Advanced RAG	Multi-hop reasoning, knowledge graphs, Naive vs Advanced vs Agentic RAG, cost optimisation, guardrails, Corrective RAG, Graph RAG.	<i>LangChain · Azure OpenAI</i>
11	Agentic AI Foundations	Agent lifecycle, tool use, API connections, router vs autonomous architectures, single vs multi-agent systems, state, failure handling.	<i>LangChain · CrewAI · AutoGen</i>
12	Decision Making, Planning & MCP	Goal setting, hierarchical vs reactive planning, MCP fundamentals vs OpenAI function calling, MCP setup, IDE agents, multi-tool orchestration.	<i>Anthropic Claude · AutoGen</i>
13	Custom Copilots & Multi-Modal Case Study	Custom copilots with Azure AI Studio. Project: Multi-modal travel planner - 7 agents (User, Planner, Vision, RAG, Speech, Summariser, Weather).	<i>Azure AI Studio · CrewAI</i>
14	Memory & Knowledge Retrieval in Agents	Memory architectures (short/long-term, vector vs symbolic, episodic vs semantic), compression, Agentic RAG, customer support case study.	<i>LangChain · ChromaDB</i>
15	LangGraph Orchestration & Multi-Agent Systems	LangGraph state machines, agent graphs, CrewAI orchestration, comparing LangGraph vs CrewAI vs Flowise, resilient workflows.	<i>LangGraph · CrewAI · Flowise</i>
16	Learning, Adaptation & Deployment	RL fundamentals, RLHF, reward systems. App structure (API+Worker+VectorDB), hosting, latency, observability with LangSmith.	<i>Streamlit · HuggingFace · LangSmith</i>
17	Advanced Multi-Agent Orchestration	Build a full LangGraph multi-agent system. Router to Planner to parallel Research+Analysis to Synthesis. Deploy end-to-end.	<i>LangGraph · FastAPI</i>

#	Session	What You Build / Learn	Tools
18	Evaluation, Debugging & Ethics	Agent evaluation metrics, logging, tracing, LLM-as-a-judge, hallucination detection, guardrails, failure modes, responsible AI.	<i>LangSmith</i>
19	Capstone - Build Phase	AI Financial Advisor: Data to Analysis to Risk to Advisory to QA agents. Full LangGraph orchestration, evaluation, FastAPI + Streamlit.	<i>LangGraph · FastAPI · Streamlit</i>
20	Capstone - Present, Deploy & Graduate	Final deployment, live Q&A, peer showcase. Each participant presents. Evaluation against production criteria. Certificate awarded.	<i>All Tools · Certificate</i>

Frameworks & Tools

Python · LangChain · LangGraph · CrewAI · AutoGen · Flowise · Azure OpenAI · Azure AI Studio · HuggingFace · ChromaDB · FAISS · FastAPI · Streamlit · LangSmith · Anthropic Claude · TensorFlow · NLTK · SpaCy